ACTIVITY 2.1 · UNIT 2 — MECHANISMS & MOTION

Observing Mechanisms

Student Edition — VEX EXP · C++ in VS Code

Name: ________________________________ **Date:** ____________ **Period:** ________

Introduction

A **mechanism** is a device that takes one kind of motion or force and changes it into another — its direction, speed, torque, or type. Motion comes in four types: **rotary** (turning), **linear** (straight line), **reciprocating** (back-and-forth in a line), and **oscillating** (swinging in an arc), and the six simple machines combine into the mechanisms all around us. The mechanism you'll use most this semester is the **gear train** — and on your EXP robot, whose motor speed is fixed, the gear ratio is the only speed-and-strength dial you have. In this activity you'll observe mechanisms on everyday objects, then build a gear train and feel the speed–torque trade with your own hands. Read Chapter 3 in the textbook for the full explanation.

Learning Objectives

- Classify motion as rotary, linear, reciprocating, or oscillating, and spot the six simple machines inside everyday mechanisms.
- Identify the drive and driven gears in a gear train and explain why meshed gears turn in opposite directions.
- Calculate a gear ratio from tooth counts and predict what happens to speed and torque.
- Choose and justify a gear ratio for an EXP mechanism (drivetrain vs. arm) using the speed–torque trade.

Key Ideas (quick reference)

- **Four motions:** rotary (wheel), linear (drawer), reciprocating (saw blade), oscillating (swing). Mechanisms convert one into another — a rack and pinion turns rotary into linear.
- **Drive vs. driven:** the drive gear is the one the motor turns (input); the driven gear is the output. Meshed gears always turn in opposite directions. Find the drive gear FIRST — it's where most wrong answers come from.
- **Gear ratio = driven teeth : drive teeth.** 24T driving 60T → 60:24 = 2.5:1 — the motor turns 2½ times per output turn. Bigger than 1 = slower and stronger; less than 1 = faster and weaker.
- **The trade:** torque up means speed down, always. More torque: smaller drive gear or bigger driven gear. Memory hook: slow down to power up.
- **Your EXP numbers:** the Smart Motor (5.5W) is fixed at 200 RPM and 0.5 N·m — no swappable cartridges (that's V5). Your kit's gears are 24T, 36T, 48T, and 60T; external gearing is your only ratio tool. 24T→60T gives an arm 2.5× the torque at 80 RPM.

Equipment & Materials

- VEX EXP parts: 24T/36T/48T/60T gears, shafts, bearing flats, shaft collars, flat bars or C-channels
- Everyday objects with mechanisms (scissors, eggbeater, can opener, bicycle)
- Engineering notebook

Procedure

1. **See the four motions.** Watch the demonstration of rotary, linear, reciprocating, and oscillating motion; copy the four into your notebook with one example of your own per type.

2. **Observation stations.** At each station, record the input motion, the mechanism (and the simple machines inside it), and the output motion in the table below.
3. **Build and feel the trade.** Mount a 24T gear meshed with a 60T gear on a flat bar (bearing flats behind each shaft, collars to hold them). Turn the 24T by hand while your partner gently resists the 60T shaft — feel how strong the slow shaft is. Then drive the 60T and feel the 24T spin fast but give up easily. Write one sentence about what you felt.
4. **Complete the gear-ratio practice table** below — the first row is worked for you.
5. **Class discussion:** which simple machines appeared at each station, and where did motion change type?

Observation Stations

Station / object	Input motion	Mechanism & simple machines	Output motion
Eggbeater (example)	*Rotary (hand crank)*	*Gear train — wheel & axle + gears; big drive gear, small driven gears → speeds up*	*Rotary (faster)*

Gear-Ratio Practice Table

Ratio = **driven teeth : drive teeth**. Bigger than 1 → slower and stronger. The motor spins 200 RPM.

Drive (input) teeth	Driven (output) teeth	Gear ratio	What happens to speed & torque?
24	48	*48:24 = 2:1*	*Drive turns twice per output turn — speed halves (200 → 100 RPM), torque doubles.*
36	60		
48	48		
60	24		
24	60		

Conclusion Questions

Answer in complete sentences.

1. Give an example of a mechanism that converts rotary motion into linear motion.
2. Why do engineers care about the type of motion a mechanism produces?
3. A 24-tooth gear on the motor drives a 60-tooth gear on a wheel. What is the gear ratio, what is the output speed if the motor spins 200 RPM, and which shaft has more torque?
4. Fill in the blanks: As the torque of a gear train increases, the speed ________. To increase torque, you should ________ the size of the drive gear or ________ the size of the driven gear.
5. Pedaling a bike up a steep hill, you shift to a low gear. Which way are you trading speed and torque, and why does it help?
6. Your robot's arm motor whines and quits when lifting a full water bottle. Without a bigger motor, what's the fix — and what will you give up?
